VG-100 Sensor Module Assembly Manual

Document VG-100-AM, revision 3. This is a mock product and manual, written for the Vigil demonstration.

Revision	Change
1	First release
2	Added cable routing through notch A
3	Added unit Rev B with a pre-fitted gasket

Read chapter 4 at the bench. Chapters 3, 5 and 6 hold the values and rules that chapter 4 depends on.

1 Scope

1.1 Purpose

This manual defines how to assemble, inspect and rework the VG-100 sensor module at a manual workbench. It applies to every operator and every unit.

1.2 Unit revisions

Two unit revisions are in production. Rev A ships as loose parts, including a separate gasket. Rev B ships with the gasket pre-fitted to the tray. The revision is printed on the unit traveller and in the unit serial label.

1.3 Order of work

Steps must be done in the published order unless a note in this manual allows otherwise. An operator must not change the order. A step is complete only when its check passes.

2 Safety and handling

2.1 Electrostatic protection

Wear a grounded wrist strap at the bench. Hold the module by its case and never by its connector.

2.2 Handling

Keep the tray in the taped fixture position. Keep each part in its labelled bin until it is needed. Do not stack parts in the tray.

2.3 Stop rules

Stop work when the system calls a hold, when a part is damaged, or when a check fails. Do not continue until the fault is corrected or a supervisor releases the unit.

3 Parts, tools and specifications

3.1 Parts list

Item	Part	Bin	Quantity
1	Gasket	A	1
2	Module	B	1
3	Cable	C	1
4	Cover	D	1
5	Clip	E	2

The tray stays in the fixture and is not a bin item.

3.2 Tools

No tool is needed for the demonstration build. Production units need a calibrated torque driver and an inspection lamp.

3.3 Fastener specifications

Fastener	Unit	Value
Binder clip, 19 mm	Demonstration units	Fully closed on the cover edge
M3 cover screw	Production units	0.60 N m, plus or minus 0.05 N m
M2 connector screw	Production units	0.25 N m, plus or minus 0.03 N m

Fasten the left side before the right side, so the cover seats evenly.

3.4 Wiring paths

| Cable | From | Path | To |
| C1 | Module output port | Through tray notch A | External connector |

Notch B is reserved for the service harness. Cable C1 must never pass through notch B, because the notch edge can damage the insulation when the cover closes.

4 Assembly procedure

Complete the steps in order. Take each part from the bin named in the step.

4.1 Lay the gasket in the tray

Take the gasket from bin A. Lay it flat inside the tray, touching the walls all the way round. Check that no part of the gasket is twisted.

Note 4.1: Rev B units ship with the gasket pre-fitted. On a Rev B unit, skip step 4.1 and start at step 4.2. This is not a sequence error.

4.2 Seat the module on the gasket

Take the module from bin B. Seat it inside the gasket with the output port facing notch A. The gasket must already be in place on Rev A units, because a module fitted without it can short against the tray.

4.3 Plug in the cable and route it through notch A

Take cable C1 from bin C. Plug it into the module output port. Route the cable out of the tray through notch A. The cable must not pass through notch B.

4.4 Fit the cover

Take the cover from bin D. Fit it onto the tray with the cable still routed through notch A. Do not trap the cable under the cover edge.

4.5 Fasten two clips, left then right

Take two clips from bin E. Fasten the left clip first and the right clip second. Both clips must be fully closed on the cover edge.

5 Inspection

5.1 In-process checks

Check each step before starting the next one, with hands clear of the tray. A step that fails its check must be corrected before work continues.

5.2 Final inspection

A finished unit shows the cover seated, two clips fastened, and cable C1 leaving through notch A.

5.3 Acceptance criteria

Reject a unit if the gasket is missing on Rev A, if the cable passes through notch B, if the cover is not seated, or if fewer than two clips are fastened.

6 Rework

6.1 Sequence errors

If a part is installed out of order, remove every part installed after the missed step. Then complete the missed step and continue in order. On Rev A, a module fitted before the gasket must be removed and inspected before reuse.

6.2 Cable routing errors

If the cable passes through notch B, remove the cover if fitted, withdraw the cable, inspect the insulation, and route it again through notch A.

6.3 Damaged parts

Quarantine any damaged part and record it. Do not reuse a damaged gasket or cable.

6.4 Repeat faults

If the same fault occurs three times at one step, stop the line and raise the step with the process engineer. Repeat faults point to the method or the instruction, not the operator.

7 Records

7.1 Unit traveller

Record the unit serial, revision, completed steps, faults and rework for every unit.

7.2 Incident records

Each confirmed fault needs an incident record that names the step, the fault, the rework carried out and the manual section that applies.

7.3 Retention

Keep records for the life of the product. Evidence images follow the same retention period and show the bench, not the operator's face.